Calibration results
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Camera-system parameters:
cam0 (/flir/blackfly/cam0/image_raw):
type: <class 'aslam_cv.libaslam_cv_python.DistortedPinholeCameraGeometry'>
distortion: [-0.39444064 0.17985589 0.00041448 -0.00020059] +- [0.00525062 0.01037597 0.00071908 0.00046127]
projection: [1006.02019963 1007.76916459 707.12127394 544.55282104] +- [0.38006568 0.4483473 0.04379318 0.71127832]
reprojection error: [0.000206, 0.000023] +- [0.124858, 0.100794]

cam1 (/zed2i/zed_node/left/image_rect_color):
type: <class 'aslam_cv.libaslam_cv_python.DistortedPinholeCameraGeometry'>
distortion: [0.00426241 -0.03063146 0.00005567 -0.00079751] +- [0.00700836 0.01287261 0.00096588 0.00068436]
projection: [527.55910599 528.56245799 647.197501   357.24769353] +- [0.2384122 0.20945034 0.05748375 0.84066972]
reprojection error: [0.000019, -0.000107] +- [0.225131, 0.189757]

cam2 (/zed2i/zed_node/right/image_rect_color):
type: <class 'aslam_cv.libaslam_cv_python.DistortedPinholeCameraGeometry'>
distortion: [0.00561685 -0.03723299 0.00017837 0.0000282] +- [0.00811179 0.01687534 0.00101465 0.00070963]
projection: [528.91033589 530.23888646 648.5912695 356.37488151] +- [0.21526847 0.24809643 0.16797954 0.87452533]
reprojection error: [0.000032, -0.000081] +- [0.208738, 0.189294]

cam3 (/flir/adk/image_thermal):
type: <class 'aslam_cv.libaslam_cv_python.DistortedPinholeCameraGeometry'>
distortion: [-0.45166823 0.2208176 -0.00011617 0.00035483] +- [0.00621688 0.01333747 0.00081382 0.00049483]
projection: [513.82316698 514.77325016 313.69598282 261.05264227] +- [0.21258681 0.25669204 0.08087125 0.92726497]
reprojection error: [-0.000085, -0.000038] +- [0.124986, 0.120911]

cam4 (/mapir_cam/image_raw):
type: <class 'aslam_cv.libaslam_cv_python.DistortedPinholeCameraGeometry'>
distortion: [0.04264083 -0.05721183 -0.00017161 -0.00088331] +- [0.00699568 0.0134804 0.00099534 0.00067121]
projection: [703.63178135 704.96034274 613.30439553 376.0574845] +- [0.38016438 0.26460237 0.18842314 0.79998821]
reprojection error: [-0.000217, -0.000019] +- [0.267507, 0.371298]

baseline T_1_0:
q: [0.00197854 -0.00806297 0.00157213 0.9999643] +- [0.00177413 0.00072472 0.00018973]

t: [0.06627231 -0.09569616 0.01543099] +- [0.00110975 0.00113314 0.00100817]

baseline T_2_1:
q: [0.00133201 0.00131534 -0.00002572 0.99999825] +- [0.00216101 0.00108292 0.00024127]
t: [-0.11982032 0.00034092 0.00429416] +- [0.00154526 0.00156119 0.00129617]

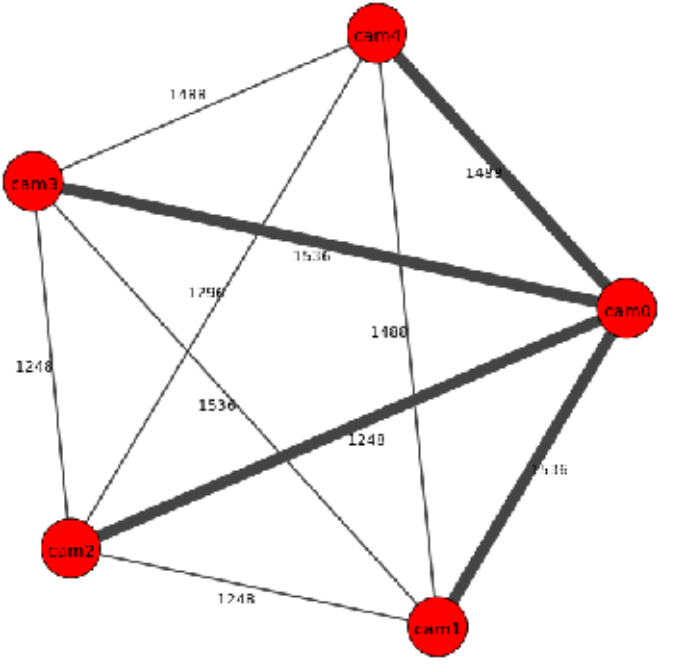
baseline T_3_2:
q: [0.00256096 -0.00318425 0.00587283 0.99997441] +- [0.00243142 0.00104766 0.00022549]
t: [-0.02410604 0.09502741 -0.02064692] +- [0.00159332 0.00160141 0.00131274]

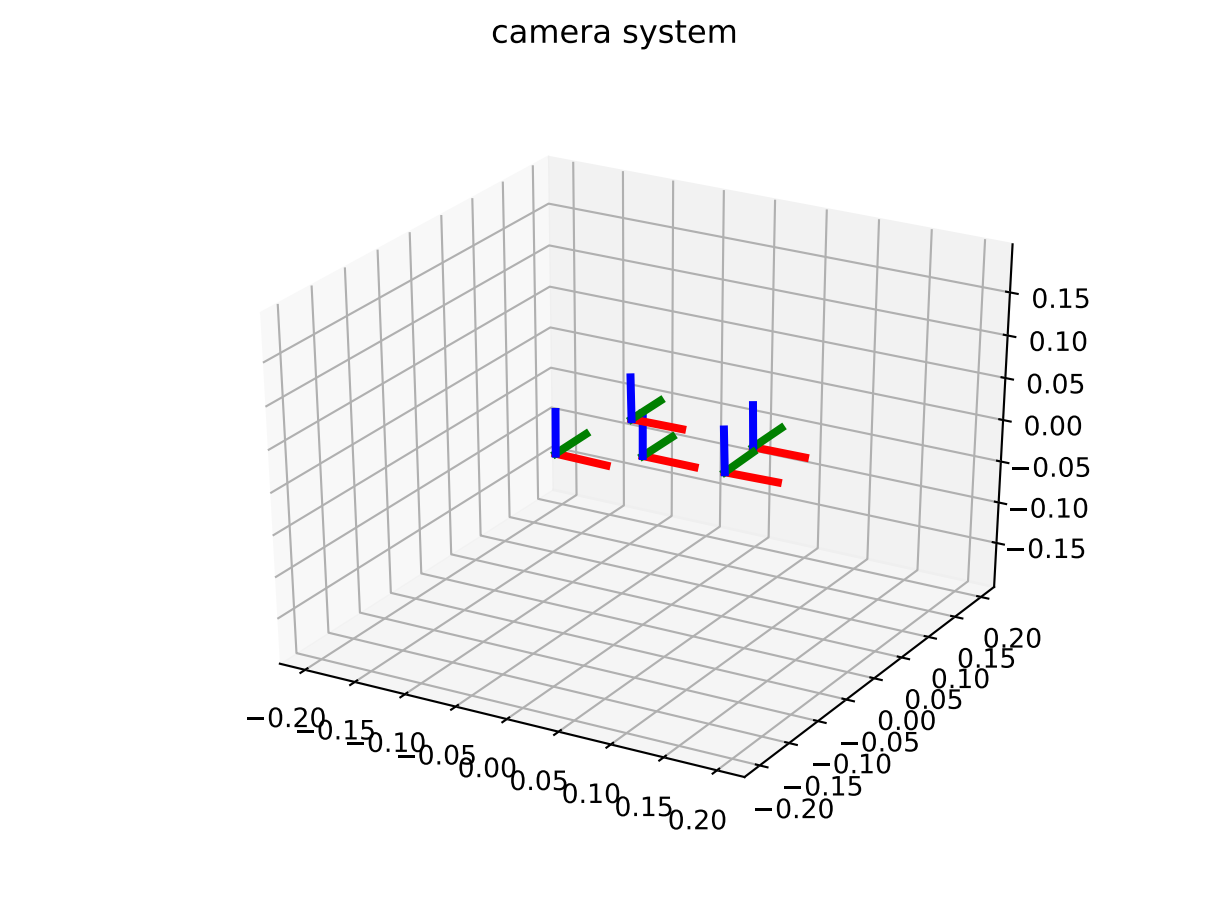
baseline T_4_3:
q: [0.00284995 0.01859189 -0.00939472 0.99977895] +- [0.00213565 0.00101437 0.00022708]
t: [0.16008285 0.00776694 0.01973759] +- [0.00134548 0.00123317 0.00111227]

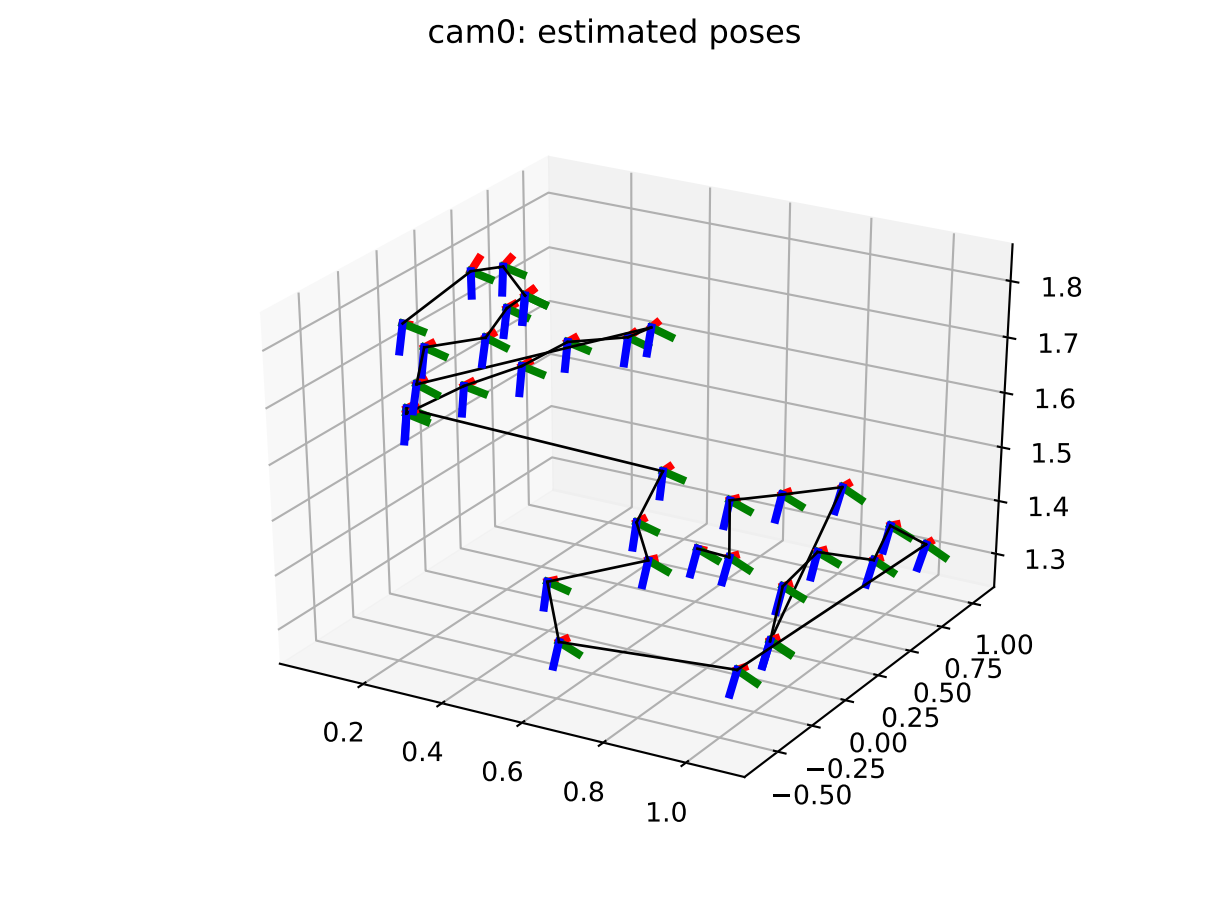
Target configuration
====================

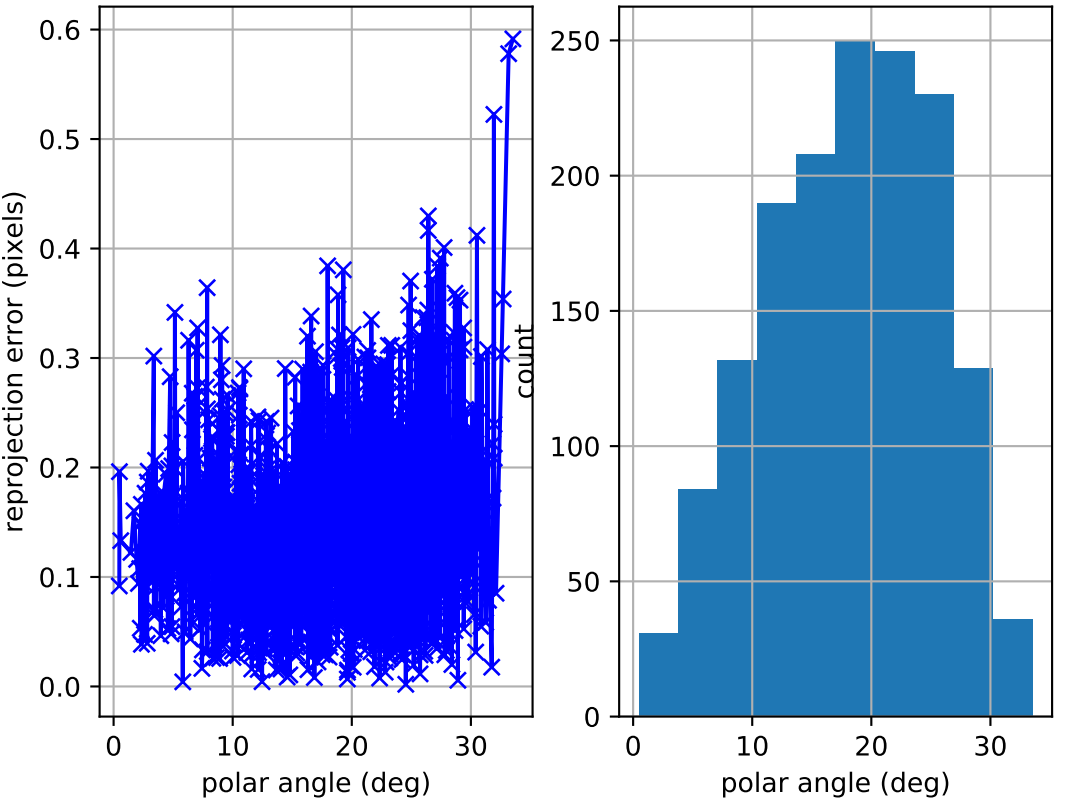
Type: checkerboard
Rows
Count: 8
Distance: 0.06 [m]
Cols
Count: 6
Distance: 0.06 [m]

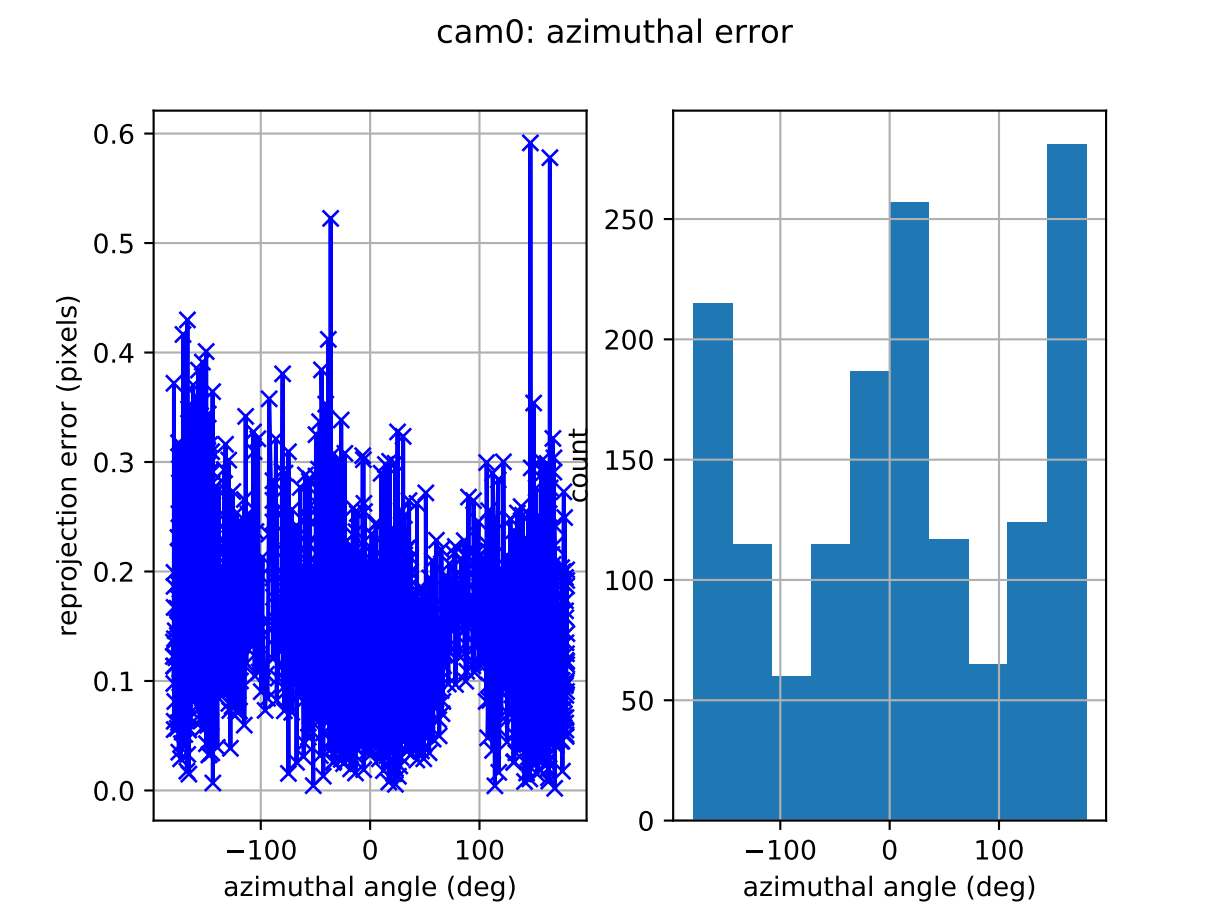
Inter-camera observations graph (edge weight=#mutual obs.)

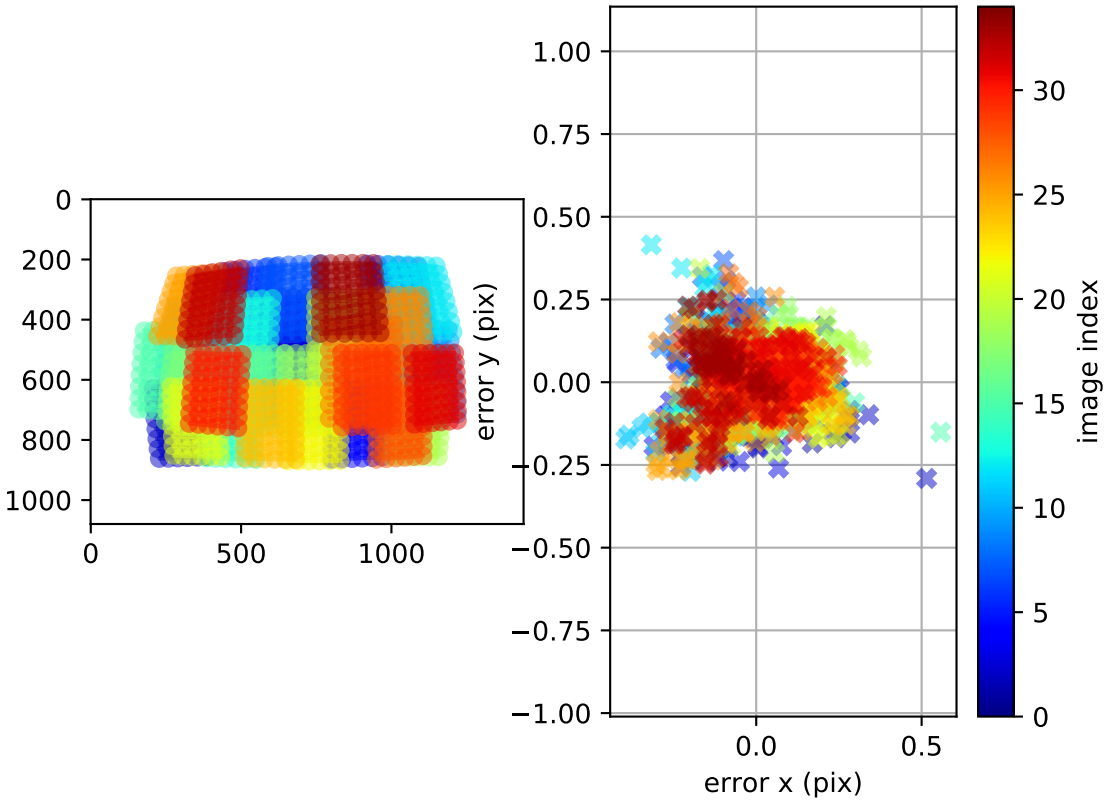


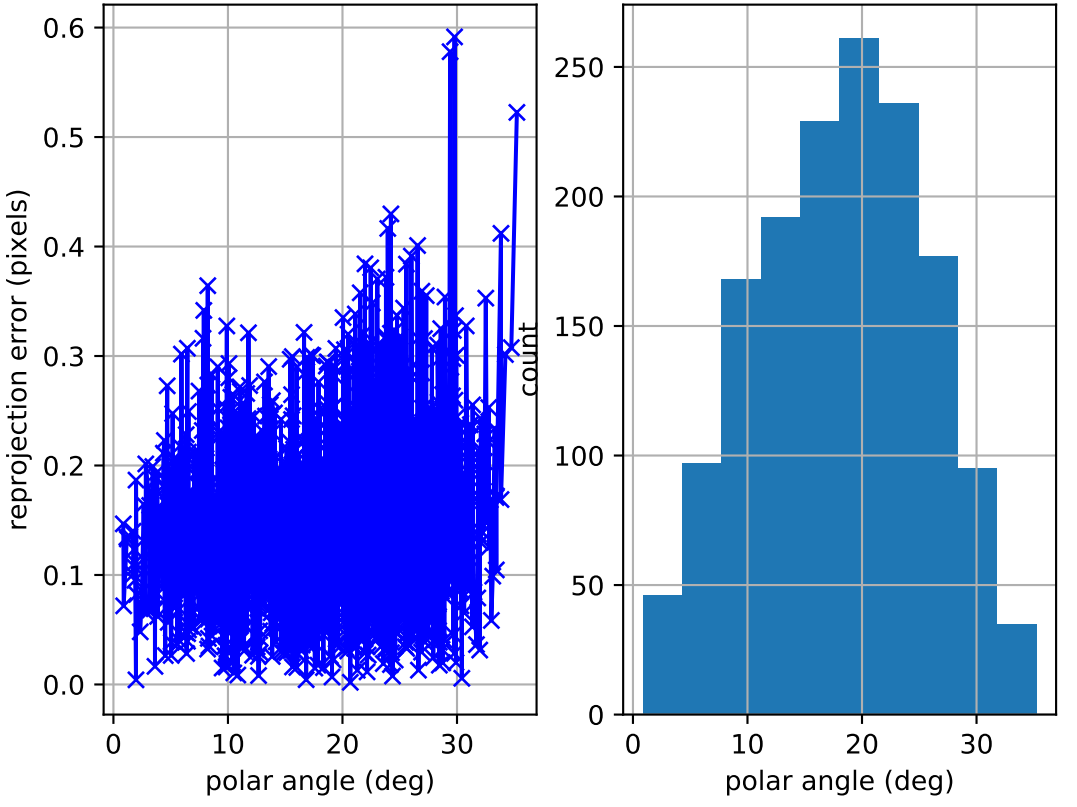

camera system
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20

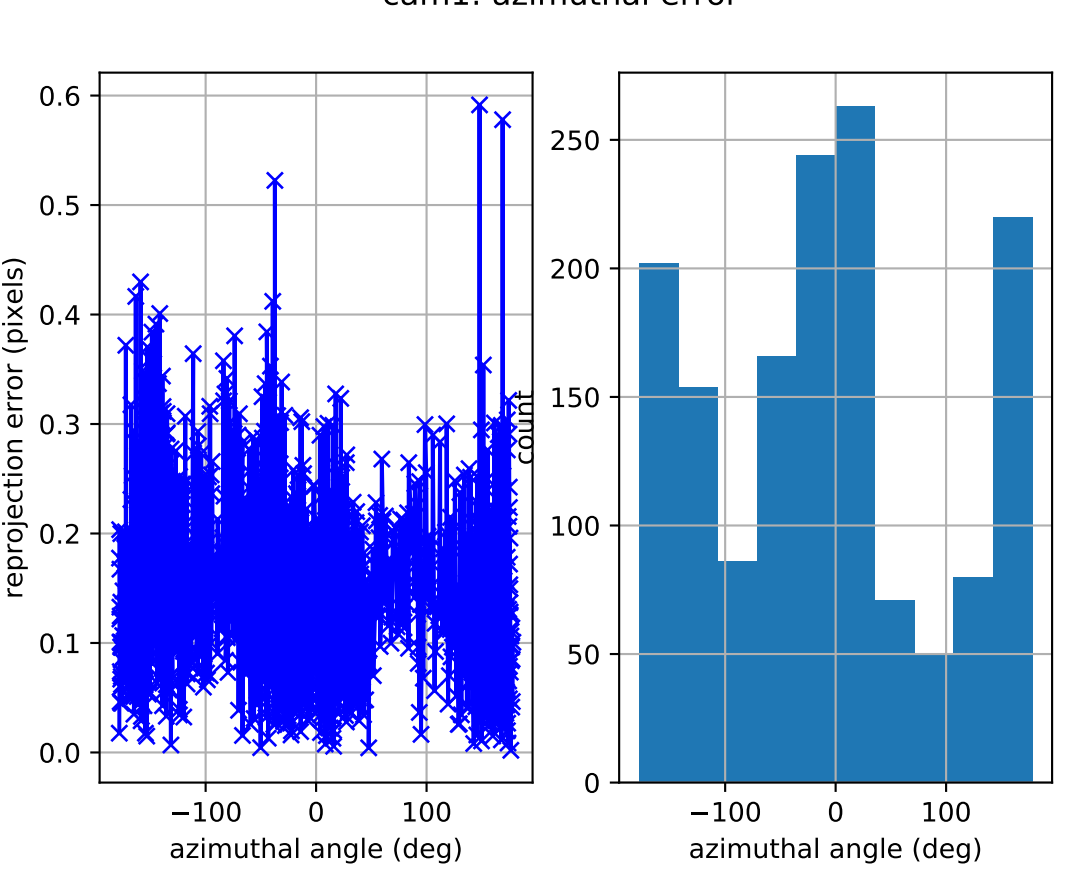

cam0: estimated poses
1.8
1.7
1.6
1.5
1.4
1.3
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0.2
0.4
0.6
0.8
1.0

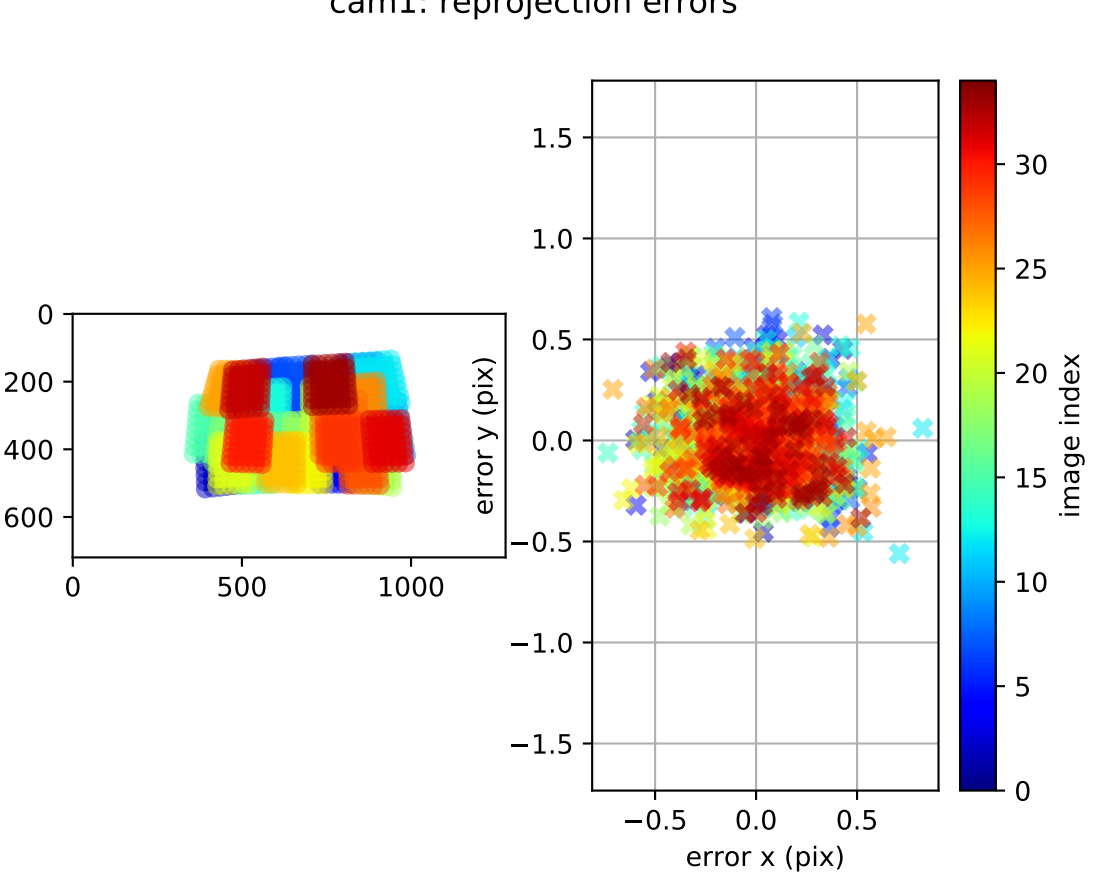

cam0: polar error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
polar angle (deg)
count
0
50
100
150
200
250
0
10
20
30
polar angle (deg)

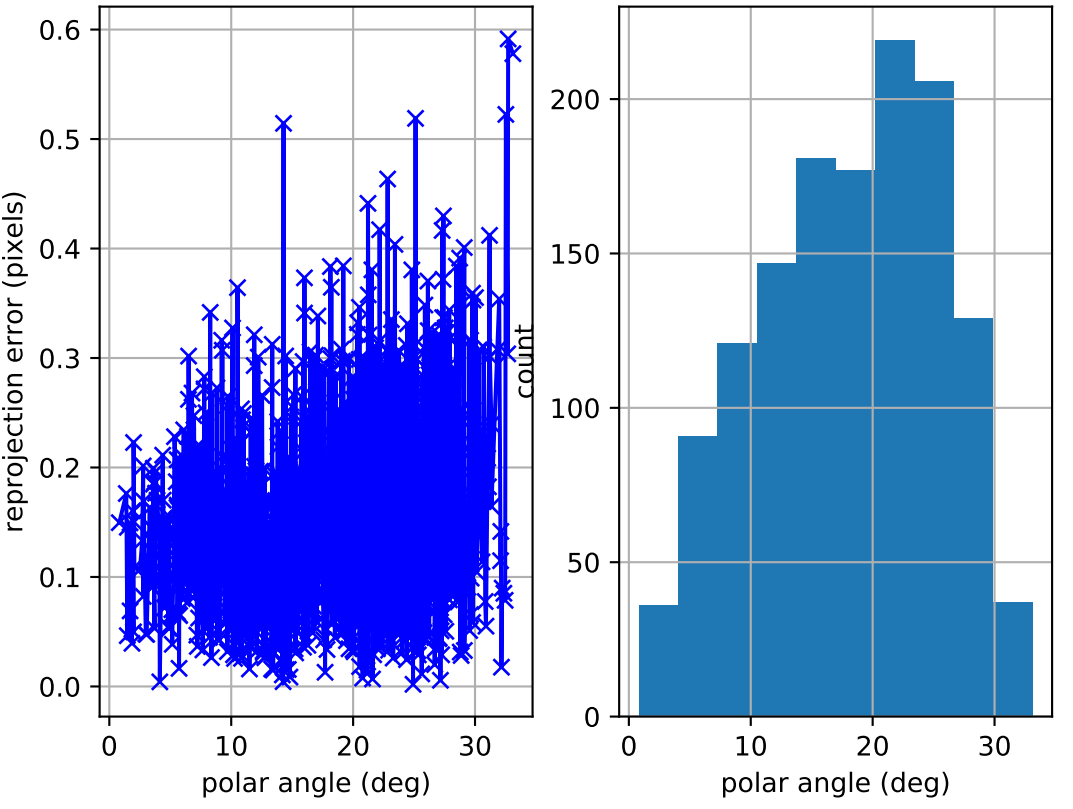

cam0: azimuthal error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
−100
0
100
azimuthal angle (deg)
count
0
50
100
150
200
250
−100
0
100
azimuthal angle (deg)

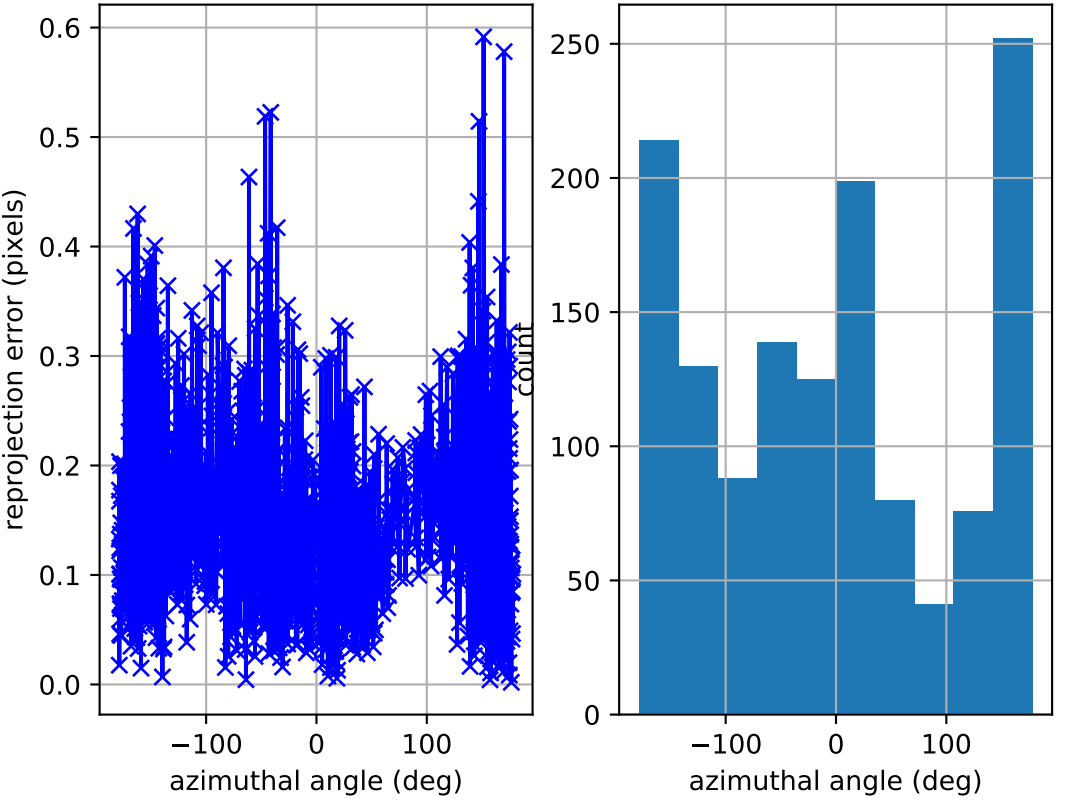

cam0: reprojection errors
0
200
400
600
800
1000
0
500
1000
error y (pix)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
0.5
error x (pix)
30
25
20
15
10
5
0
image index

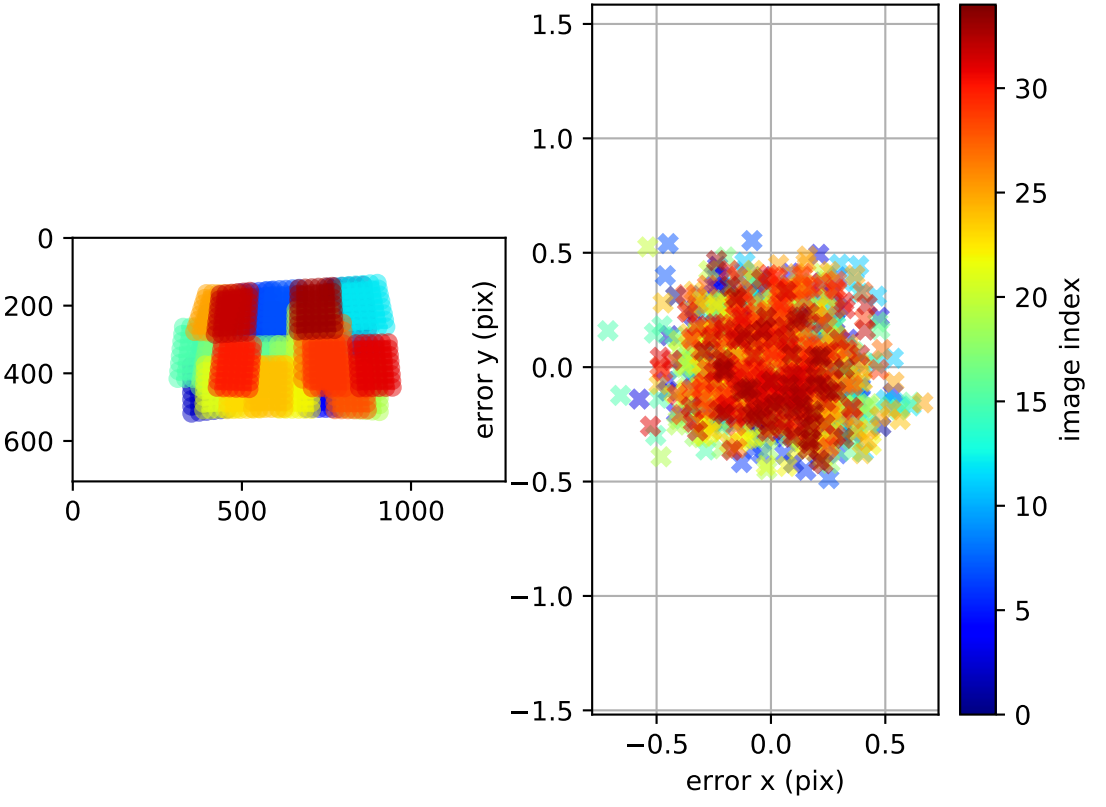

cam1: polar error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
polar angle (deg)
count
0
50
100
150
200
250
0
10
20
30
polar angle (deg)

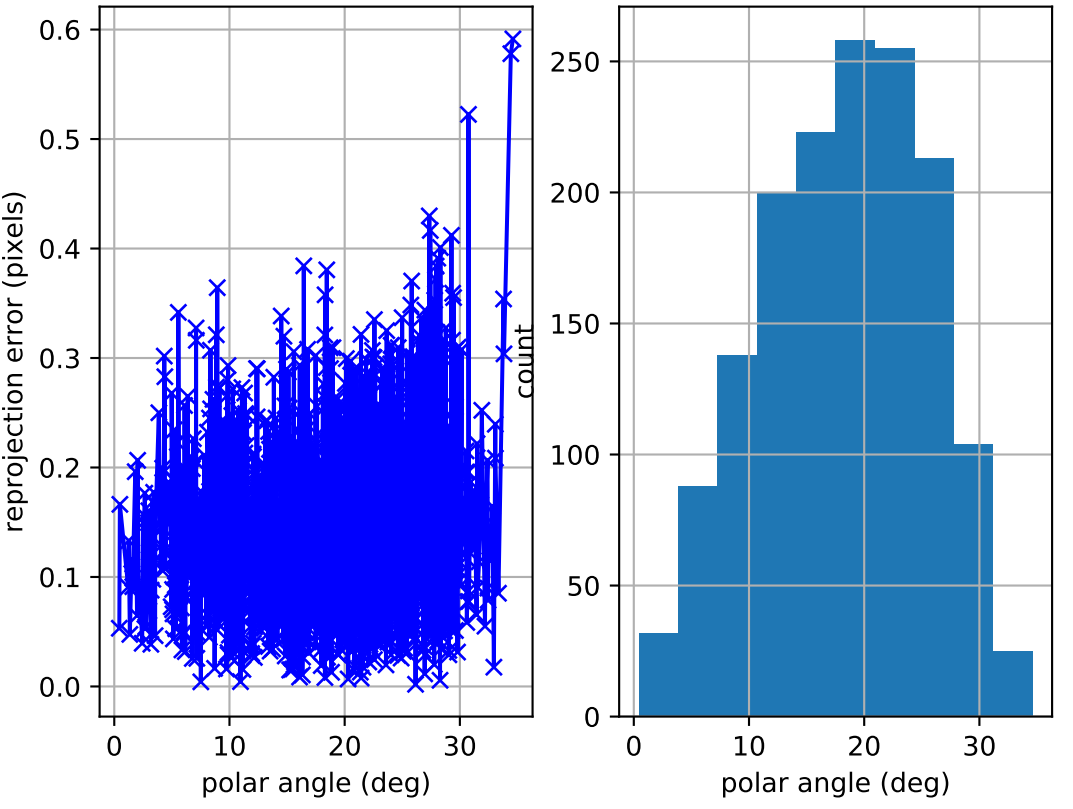

cam1: azimuthal error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
−100
0
100
azimuthal angle (deg)
count
0
50
100
150
200
250
−100
0
100
azimuthal angle (deg)

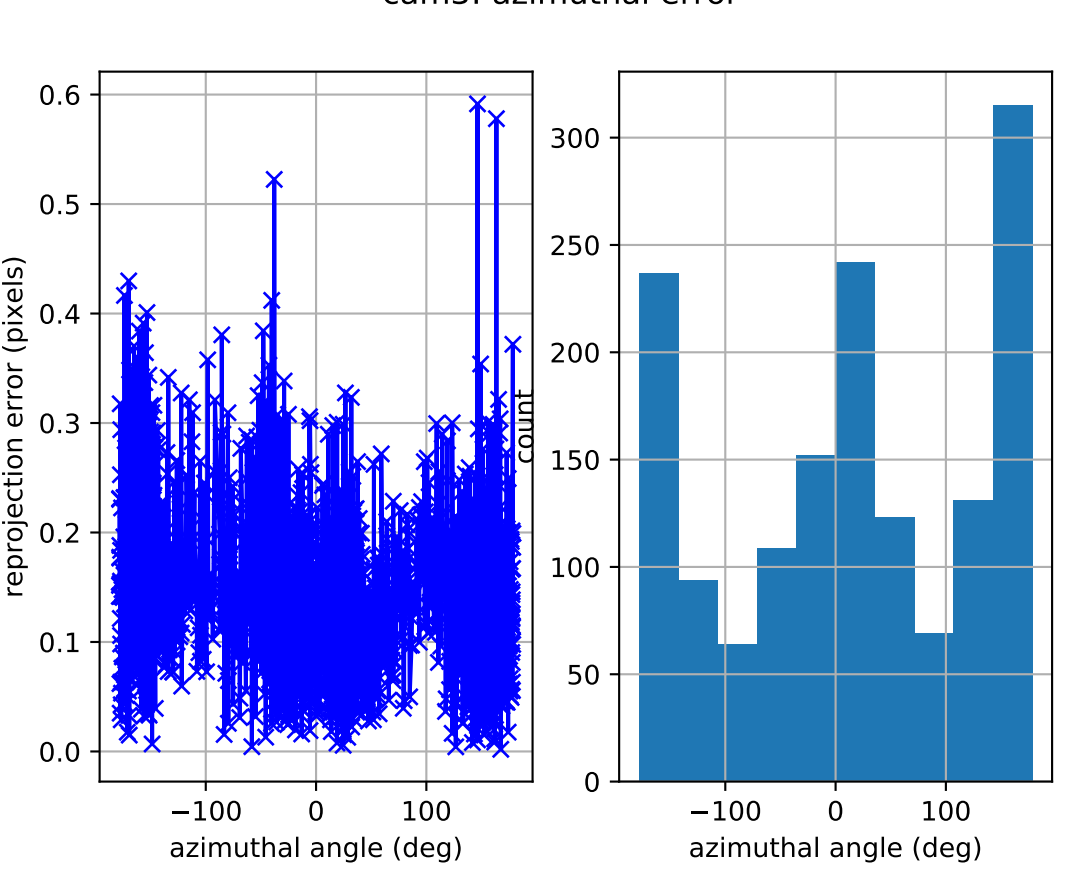

cam1: reprojection errors
0
200
400
600
0
500
1000
error y (pix)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.0
0.5
error x (pix)
30
25
20
15
10
5
0
image index

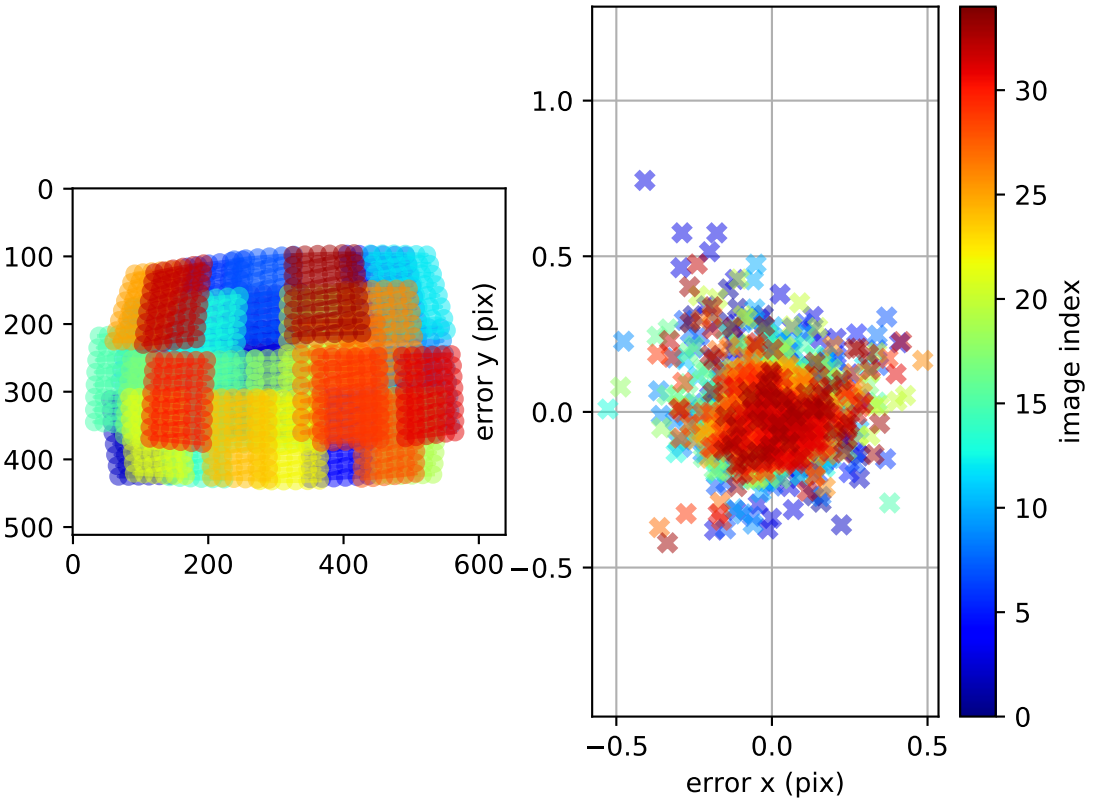

cam2: polar error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
polar angle (deg)
count
0
50
100
150
200
0
10
20
30
polar angle (deg)

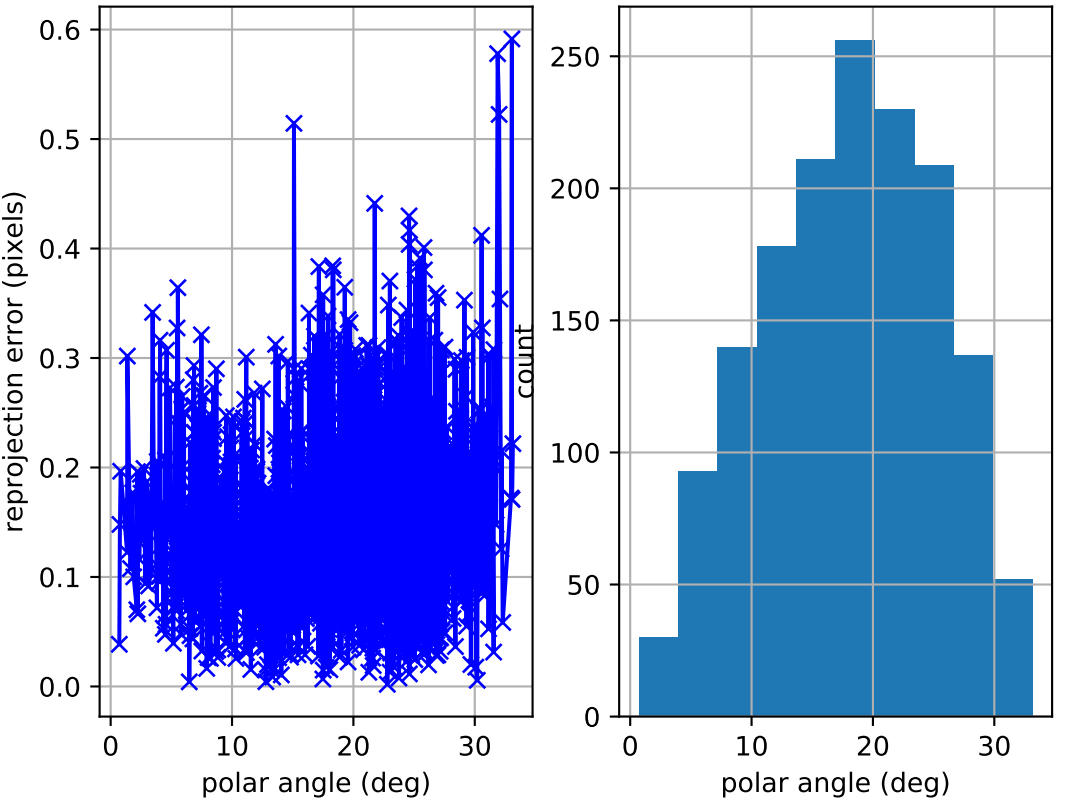

cam2: azimuthal error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
−100
0
100
azimuthal angle (deg)
count
0
50
100
150
200
250
−100
0
100
azimuthal angle (deg)

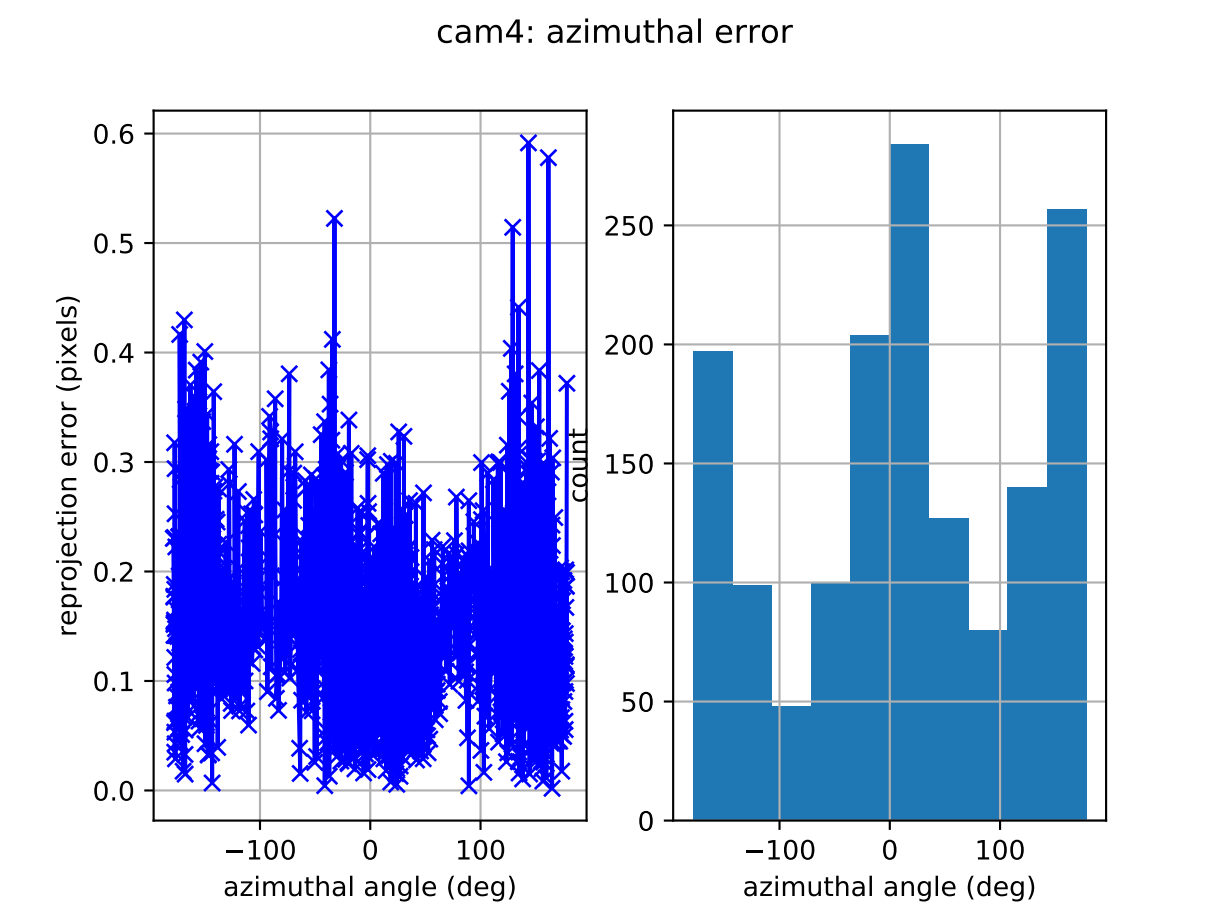

cam2: reprojection errors
0
200
400
600
0
500
1000
error y (pix)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.0
0.5
error x (pix)
30
25
20
15
10
5
0
image index

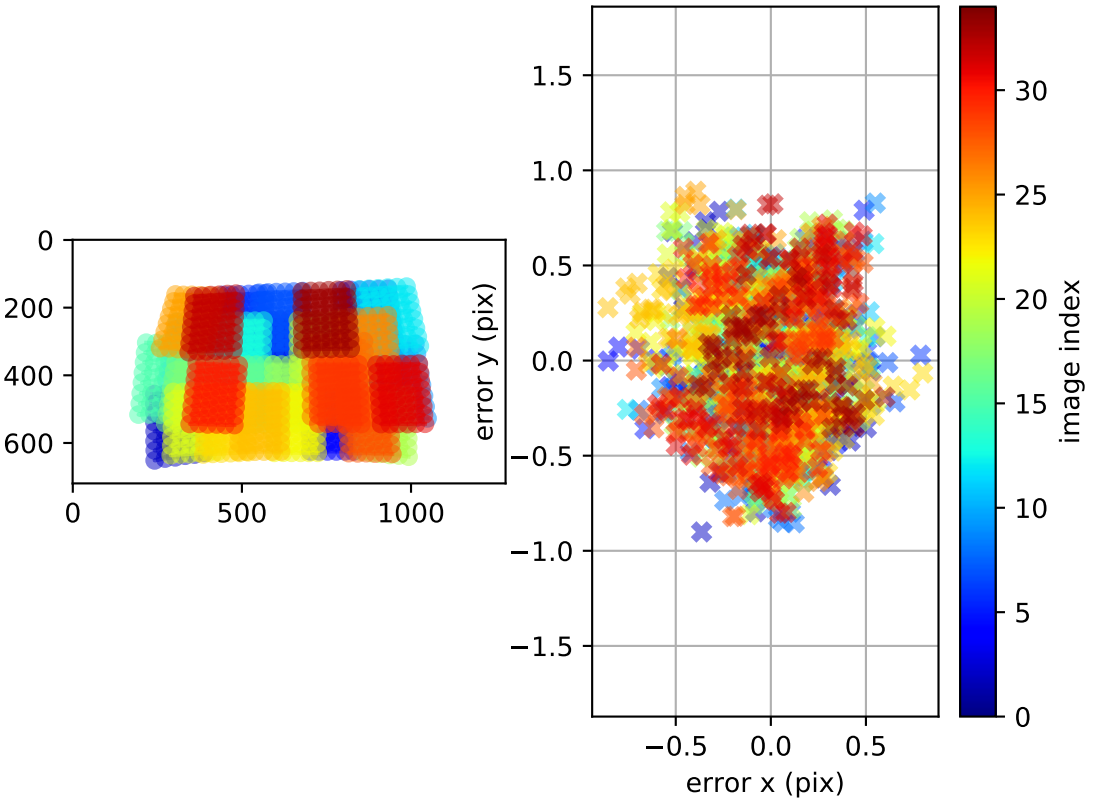

cam3: polar error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
polar angle (deg)
count
0
50
100
150
200
250
0
10
20
30
polar angle (deg)


cam3: azimuthal error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
−100
0
100
azimuthal angle (deg)
count
0
50
100
150
200
250
300
−100
0
100
azimuthal angle (deg)


cam3: reprojection errors
0
100
200
300
400
500
0
200
400
600
error y (pix)
1.0
0.5
0.0
−0.5
−0.5
0.0
0.5
error x (pix)
30
25
20
15
10
5
0
image index


cam4: polar error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
polar angle (deg)
count
0
50
100
150
200
250
0
10
20
30
polar angle (deg)


cam4: azimuthal error
reprojection error (pixels)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
−100
0
100
azimuthal angle (deg)
count
0
50
100
150
200
250
−100
0
100
azimuthal angle (deg)


cam4: reprojection errors
0
200
400
600
0
500
1000
error y (pix)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.0
0.5
error x (pix)
30
25
20
15
10
5
0
image index